

Miguel Angel Gonzalez Delgadillo

PROFESSIONAL SUMMARY

Senior Software Engineer / Kafka Integration Engineer with 11+ years of experience designing, optimizing, and supporting high-availability, large-scale distributed systems.

Strong expertise in Apache Kafka, Java (1.4–17), Python, Spring Boot, Kubernetes, Linux (RHEL), and event-driven architectures. Experienced in building high-throughput, low-latency data pipelines and implementing reliable messaging systems using advanced Kafka features such as idempotent producers, manual offset management, and performance tuning.

Proven experience in production-critical environments (24/7), observability, system performance optimization, and cross-team collaboration in complex telecom and media platforms.

INFORMATION

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EDUCATION

Máster 's in Computer science – *ISU Universidad – Puebla, México*

- Start date: January 2022
- End date: January 2024

Bachelor's in computer science – *ITS - Saltillo, Coahuila*

- Start date: January 2009
- End date: January 2014

LANGUAGES

English
Spanish

CERTIFICATIONS

Splunk Core Certified User
Expedition: oct. 2021
ID: 279681237

SKILLS

Languages: Java (1.4–17), Python, Bash, PHP Frameworks: Spring Boot, Flask Messaging & Streaming: Apache Kafka (Advanced Producer/Consumer Design, Idempotence, Offset Management, Lag Monitoring, Partitioning Strategies) Kafka Performance: Throughput Optimization, Latency Tuning, Batching (linger.ms, batch.size), Compression (Snappy, LZ4, Zstd) Containers & Orchestration: Docker, Kubernetes Databases: Oracle, MySQL DevOps & CI/CD: Jenkins, Harness, Git (Bitbucket) Monitoring & Observability: Splunk, AppDynamics, Kafka Metrics Operating Systems: Linux (Red Hat 6.x–8.6) APIs & Protocols: REST, SOAP, OAuth 2.0 Cloud-Native Concepts: Microservices, Scalability, High Availability

PROFESSIONAL EXPERIENCE

Innova Solutions – Senior Engineer Lead / Software Developer IV **Start Date: September 2018** **End Date: Present**

Client: Charter Communications – Spectrum

Location: Missouri and Charlotte

Start Date: September 2018

End Date: Present

Senior Java Engineer / Production & Middleware Support Platform

- Provided production support for Java-based applications (Spring Boot, Kafka) in enterprise environments, ensuring high availability and system stability.
- Monitored systems using alerts and logs, proactively identifying and resolving issues before impacting business operations.
- Performed incident management and triage, analyzing logs, metrics, and system behavior to quickly identify root causes and restore services.
- Handled HTTP-related issues and backend service failures, including troubleshooting APIs, connectivity, and response errors.
- Supported distributed systems using Apache Kafka, ensuring message reliability and troubleshooting consumer/producer issues.
- Collaborated in incident calls and cross-functional teams to resolve production outages efficiently.
- Worked with Linux (RHEL) environments to manage services, review logs, and troubleshoot infrastructure-related issues.
- Managed and tracked incidents using ticketing systems (ServiceNow, Cherwell and JIRA), ensuring proper prioritization and SLA compliance.
- Analyzed recurring issues and contributed to root cause analysis (RCA) and long-term solutions to prevent reoccurrence.
- Assisted in deployment validation and post-release monitoring to ensure application stability.
- Troubleshot Java application servers behavior (logs, memory, performance issues) in production environments.

- Designed and implemented event-driven architectures using Apache Kafka for high-throughput, low-latency systems.
- Developed and optimized Kafka producers and consumers, including:
 - - Idempotent producers to prevent duplicate messages
 - - Manual offset management to ensure reliable processing
 - - Consumer lag monitoring and performance optimization
- Tuned Kafka performance for concurrent workloads:
 - - Optimized batching using `linger.ms` and `batch.size`
 - - Balanced latency vs throughput based on system requirements
- Designed scalable data pipelines using partitioning strategies to enable parallel processing across consumer groups.
- Implemented reliable message processing patterns aligned with exactly-once semantics concepts, leveraging idempotence and controlled offset management.
- Built and maintained real-time data streaming pipelines for mission-critical systems.
- Implemented robust error-handling strategies, including retries and backoff mechanisms, to ensure system resilience.
- Monitored Kafka ecosystem using metrics such as consumer lag, throughput, and request latency via tools like Splunk and AppDynamics to identify bottlenecks and optimize performance.
- Collaborated with DevOps and infrastructure teams to ensure Kafka cluster stability, broker health, and optimal resource utilization.
- Provided production support, troubleshooting issues related to consumer lag, broker performance, and message processing delays.
- Built Spring Boot microservices integrated with Kafka to support distributed, event-driven architectures.
- Strong understanding of Kafka internals, including ISR (In-Sync Replicas), leader election, and delivery guarantees (at-least-once and exactly-once semantics).
- Experience debugging and resolving production issues such as high latency, duplicate messages, and consumer lag.
- Familiar with Kafka client tuning and performance optimization, including configurations aligned with librdkafka-based clients.
- Built Spring Boot microservices supporting distributed systems with Kafka integration.
- Provided production support, troubleshooting issues related to consumer lag, broker performance, and message processing delays.
- Designed, developed, and maintained **enterprise-grade Java applications** using **Java 8–17**, following clean code and SOLID principles.
- Led the **migration of legacy Java 1.8 applications to Java 17**, modernizing codebases and improving performance, security, and maintainability.
- Built **Spring Boot microservices** supporting high-throughput, low-latency backend systems.
- Implemented security using Spring Security with OAuth2 and JWT in a Spring Boot application built with Java 17, securing REST APIs with role- and scope-based authorization.
- Designed and implemented **event-driven architectures** using **Apache Kafka**, including:
 - **Kafka producers and consumers**
 - **Topic and partition management**
 - **Offset tracking and lag analysis**
- Developed **RESTful and SOAP APIs**, ensuring proper error handling, validation, and API contract compliance.
- Integrated **Redis caching** to improve response times and reduce database load.
- Implemented **OAuth 2.0 authentication and authorization** across distributed Java services.
- Performed **code refactoring and cleanup** on legacy Java applications to align with modern Java standards and best practices.
- Wrote and optimized **SQL queries** and integrated Java applications with **Oracle databases**.

- Created and maintained **unit and integration tests** (JUnit) to ensure application reliability and stability.
- Participated in **code reviews**, providing technical feedback and enforcing Java best practices.
- Supported Java applications running in **containerized and Kubernetes-based environments**, including:
 - Monitoring application logs and JVM metrics
 - Troubleshooting runtime issues at the pod level
- Collaborated closely with DevOps and platform teams to ensure smooth deployments and production stability.
- Provided **24/7 production support** for Java applications, including incident resolution and root cause analysis.
- Administered **Red Hat Linux environments**, including:
 - Application installations
 - Cron jobs and startup scripts
 - System hardening and operational maintenance
 - Led the **enterprise-wide migration from RHEL 6.2 to RHEL 8.6**, ensuring compatibility, stability, and security compliance.
- Directed **code refactoring and cleanup initiatives** for legacy Java 1.8 codebases.
- Actively participated in **technical interviews and onboarding** of new engineers.
- Collaborated with infrastructure and network teams to **configure and validate firewall rules** across all environments.
- Authored and maintained **operational documentation, knowledge transfer (KT), and incident reports**.
- Managed **application certificates, licensing (AppDynamics)**, and compliance-related operational tasks.
- PowerShell **automation** to reduce manual pre-deployment verification steps and operational risk.
- Strong understanding of ISR (In-Sync Replicas) and leader election
- Experience with delivery guarantees: at-least-once and exactly-once semantics
- Knowledge of Kafka trade-offs: latency vs throughput, durability vs performance
- Experience debugging production issues: high latency, duplicates, consumer lag
- Familiar with Kafka client tuning and librdkafka-based configurations

Client: Charter Communications – Spectrum

Location: Missouri and Charlotte

Start Date: October 2025

End Date: January 2026

Python Developer – Automation & Integration Projects

- Developed automated scripts to connect to Oracle databases, extract and transform data, apply record modifications, and reinsert updated records.
- Designed and developed robust Python automation frameworks to connect to Oracle databases, perform ETL-like data extraction, transformation, validation, and reinsertion, ensuring data consistency in production environments.
- Built secure SSH-based automation tools to remotely collect Linux system metrics, logs, and configuration data, followed by automated persistence into relational databases.
- Engineered log ingestion and analysis pipelines by integrating Python services with Splunk APIs, enabling proactive monitoring and faster incident detection.
- Designed cross-database reconciliation processes to retrieve data from multiple data sources, detect inconsistencies, and generate actionable discrepancy reports.
- Developed automated reporting and notification systems delivering real-time insights via Slack, Cisco Webex, and email, improving operational visibility and response times.
- Implemented event-driven alerting systems triggered by log patterns, system activity, and threshold-based conditions in Linux environments.

- Integrated Python services with Apache Kafka 2.4, including:
- Kafka consumers and producers
- Message flow validation and lag monitoring
- Real-time data streaming and processing
- Designed and developed RESTful APIs using Flask, exposing internal services and automation capabilities to other teams and systems.
- Implemented CRUD operations backed by Oracle and relational databases, following clean architecture and modular design principles.
- Built API endpoints to support automation workflows, including database operations, system health checks, and log processing.
- Implemented structured JSON input/output, HTTP status codes, and standardized error handling compliant with REST best practices.
- Integrated REST APIs with frontend dashboards, automation platforms, and messaging tools for end-to-end workflow automation.
- Conducted API testing and validation using Postman, including functional and negative test scenarios.
- Deployed Python services and APIs on Linux-based servers, validating runtime behavior through SSH-based troubleshooting and log analysis

TXM Global – Saltillo, Coahuila

Position: Software Integrator

Start date: September 2014

End Date: September 2018

Location : Saltillo Coahuila Mexico

Software Integrator

Client: Telcel México

Location: Mexico City

Start Date: September 2018

End Date: September 2016

- Development of Cartridges documentation.
- Internal Test for Cartridges in ASAP.
- Functionality Test for Cartridge in ASAP.
- ASAP cartridge integration for SOAP communication protocol.
- Design, development of some services of the cartridge, using Design Studio 3.1.3. This cartridge is integrated into the ASAP 7.0.2.
- Installation in 10 production server
- Monitoring ASAP Servers
- Batch handler customization

Software Integrator

Client: Tigo Colombia

Location: Bogota Colombia

Start Date: September 2016

End Date: September 2014

- Upgrade Stored procedures in ASAP Database
- Upgrade of Blackberry, OTA and HLR cartridge
- Create new HLR Commands on the existing cartridge
- Create a jar file to share documents with ftp connection
- Provide remote meeting with the customer to explain the operation and maintenance of ASAP
- Upgrade Documentation with the new solution
- Provide remote meeting with the customer to explain how to fix ASAP Problems
- Provide remote meeting with the customer to explain the operation and maintenance of ASAP
- Create Documentation about how to fix Web logic authentication problem
- Provide remote meeting with the customer to explain how to fix ASAP Problems
- Oracle Communications ASAP Cartridge development (Service Activation) for RIM-Blackberry, OTA, HLR provisioning.
- ASAP cartridge integration for different communication protocols (SOAP, Telnet)
- Provide onsite and remote development cartridges of the solution.
- Batch Handler customization, installation and testing.
- Script to manipulate the Batch Handler.
- Script to send notification to Tigo Web Service.
- Identify and Analyze customer requirements.
- Provide onsite and remote meeting with the customer to explain the solution.
- Provide onsite and remote meeting with the customer to explain the operation and maintenance of ASAP
- Cartridge Installation.
- Development JMS Client to send error notifications
- Network Element configuration.
- Functionality Test for Cartridge in ASAP.
- Documentation about Acceptance Test Specification and Installation Report for ASAP.
- Support with questions about the provisioning and ASAP functionality.
- Development of project documentation
- O & M documentation
- Provided support in the production and pre-production phase.
- Deployment of cartridges over ASAP production servers.
- Execution of Functional Acceptance Tests for ASAP cartridges using SOAPUI and Web logic console.
- Create database Schema with roles and permissions
- Create PL/SQL Stored Procedures
- Create PL/SQL Functions
- Create Unix script to manipulate database information
- Update existing Scripts to manipulate information

Intern

Client: TXM Global

Location: Saltillo Coahuila

Start Date: September 2014

End Date: January 2014

- Implementation the framework CakePHP
- Using MySql Database
- Using JavaScript to make some validations
- Using Ajax to make asynchronous queries
- Using Bootstrap to front end Design
- Using JQuery for JavaScript functions
- Using Moment.js to Date Time validations
- Identify and Analyze requirements.
- Support with questions about the tool functionality.
- Deployment the tool over production servers.